CAPSTONE / AI DRIVEN DATA SCIENCE

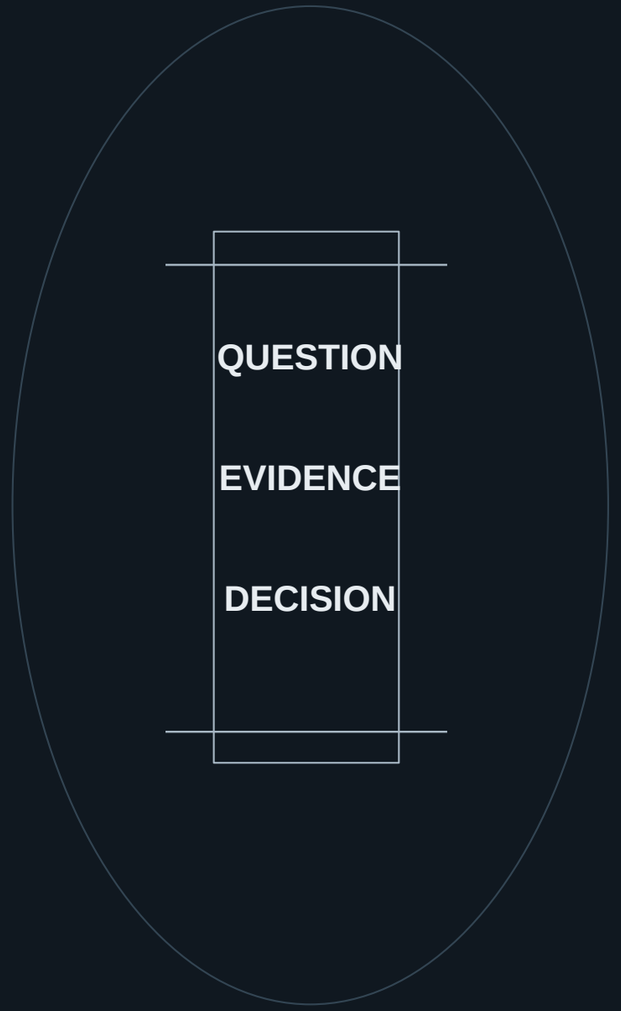
BOUNDARY

Franchise research,
with evidence attached.

A constrained AI workflow for cricket analysts and scouts.

B.R. Abishek

Capstone Project / AI Driven Data Science



THE ANALYST RETAINS THE DECISION

Start with a decision, not a dashboard.

The analyst’s problem

- Records sit across competitions and seasons.
- A ranking can hide its date window and sample size.
- A plausible XI can still fail to cover twenty overs.

The project response

- Confirm one task and its evidence scope.
- Calculate from linked match records.
- Show assumptions, constraints and exportable evidence.

The output is a research brief that a human can inspect.

Four contributions, one inspectable workflow.

01	**Evidence**	Registry-linked match data, explicit phases and dates.
02	**Planning**	Choose eleven players and assign every over together.
03	**Bounded AI**	Optional DeepSeek selection from existing fact IDs.
04	**Reproducibility**	Save the question, assumptions, versions and source hashes.

What an analyst can actually ask.

SCOUTING

Which observed death bowlers deserve video review?

XI PLANNING

Can MI cover twenty overs without Bumrah?

PLAYER EVIDENCE

What does Kohli’s selected archive sample contain?

MATCH RESEARCH

Does a research model beat its historical baseline?

Ten workspaces; a single research loop.

01 **Home**
Purpose, examples, coverage

02 **Decision desk**
Checked research briefs

03 **Scouting & comparison**
Scope, rank, compare

04 **Squad & auction scenarios**
XI, schedules, assumptions

05 **Player dossier**
Identity and archive evidence

06 **Match research**
Baseline and dated holdout

07 **Decision journal**
Save and export reasoning

08 **Evaluation & sources**
Audits, SWOT, review tasks

09 **Project guide**
Explain, install, reproduce

10 **Settings**
Private key and call allowance

Five archives. Coverage is a first-class feature.

Competition	Matches	Earliest	Latest
IPL	1,242	2008-04-18	2026-05-31
BBL	662	2011-12-16	2026-01-25
Hundred	201	2021-07-22	2026-08-16
T20I	3,592	2005-02-17	2026-10-06
SMAT	695	2016-01-02	2024-12-15

6,392 eligible matches / Frozen supplied records / No automatic live updates

From deliveries to dated player evidence.

01

Read

Five JSON ZIP archives

02

Filter

Eligible men’s matches

03

Link

Registry IDs and aliases

04

Aggregate

Player, match and phase

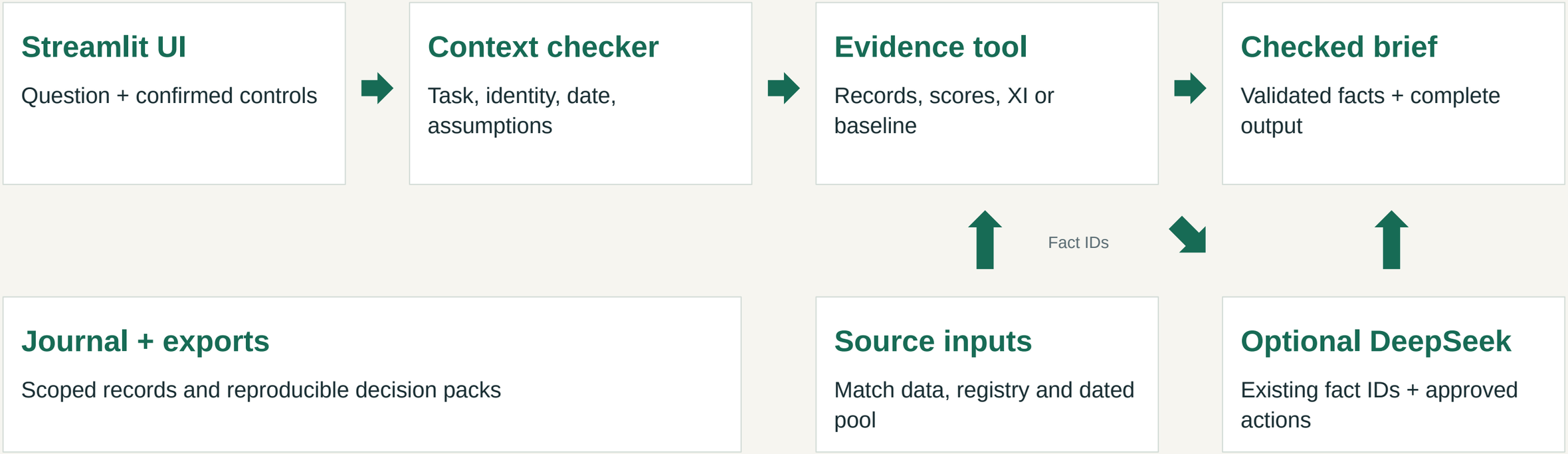
05

Verify

Independent raw checks

Super overs stay separate; voided and duplicate match IDs are excluded.

The calculation engine owns the facts.



Outputs: scoped brief, private journal and decision pack with source fingerprints

Question → evidence → check → human review.

01

Confirm

Task, identity, scope

02

Retrieve

One permitted tool

03

Calculate

Rates, scores or XI

04

Check

Rules and fact IDs

05

Review

Brief, sources, export

A conflicting request asks for clarification before calculation.

A narrow model role is a deliberate design choice.

DeepSeek can

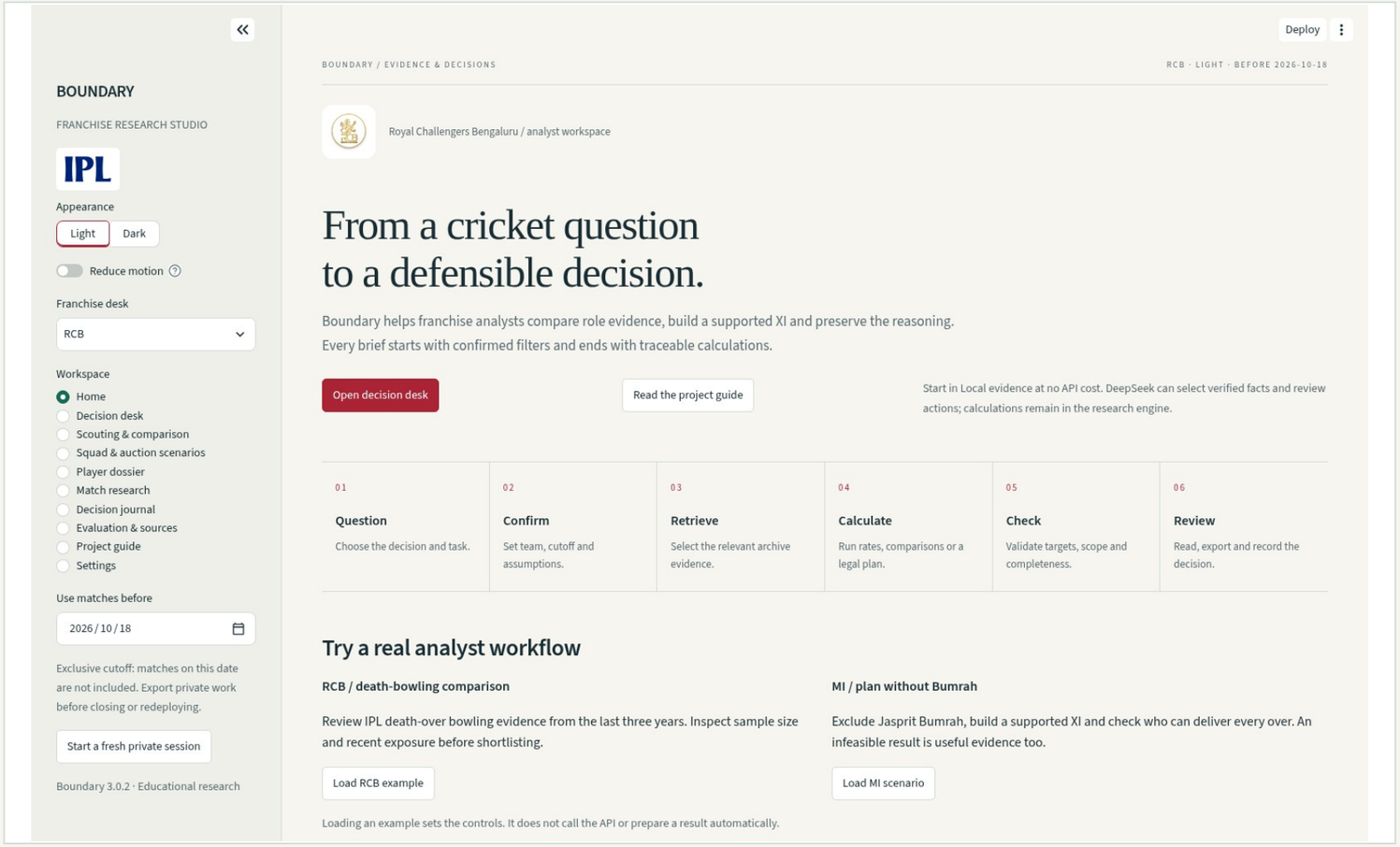
- Select useful supplied fact IDs such as F1 and F2.
- Choose approved next actions.
- Return a small validated JSON object.

The server controls

- Tools, filters, calculations and source facts.
- All eleven XI facts and the full over schedule.
- Fallback to checked local evidence.

No key is needed for Local evidence mode.

Orientation before analysis.



Actual local application · v3.0.2 · Local evidence mode

WORKFLOW

A visible path from question to review.

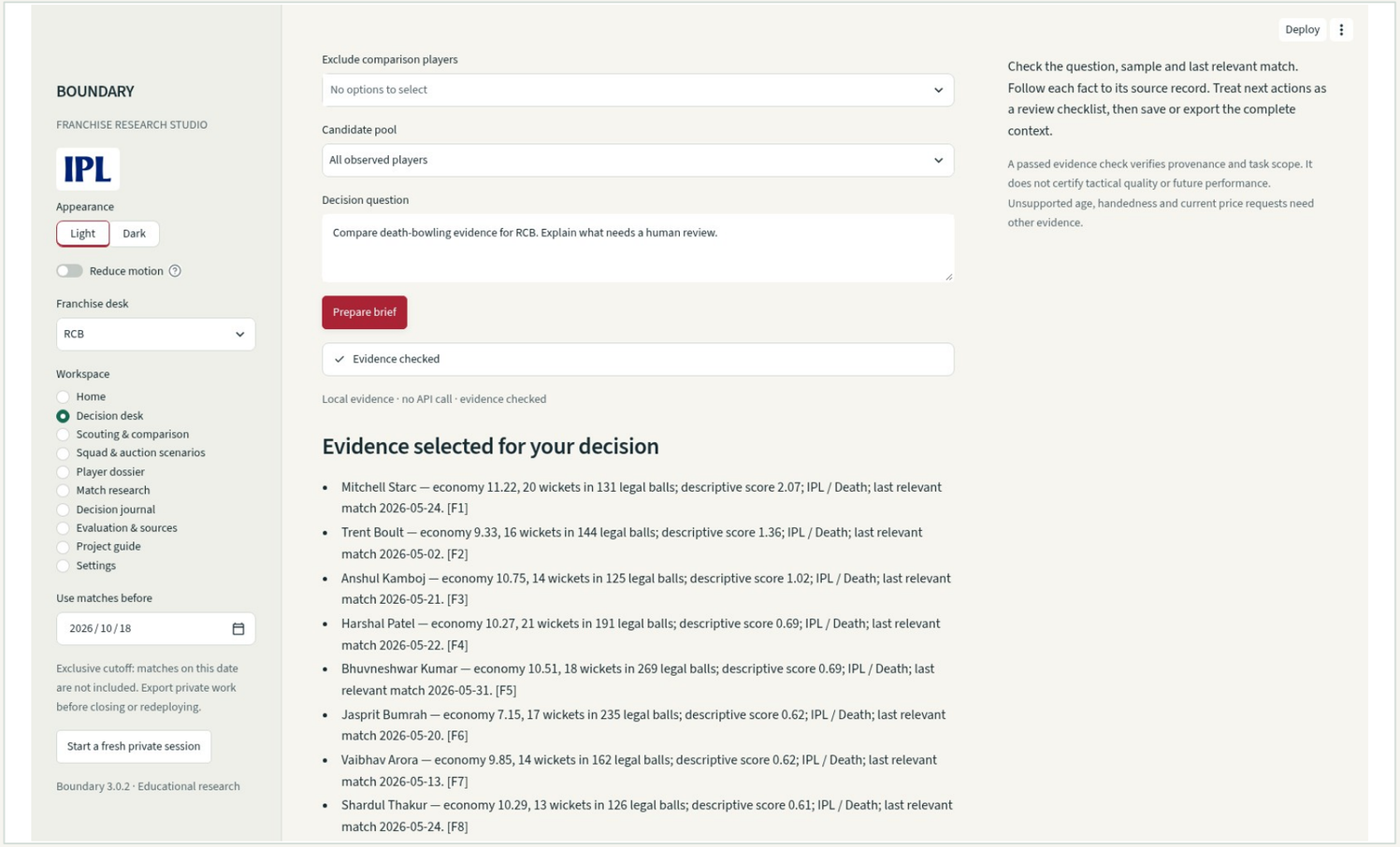
GUIDED EXAMPLES

RCB scouting and MI without Bumrah.

COVERAGE

Source endpoints and project downloads.

Prepare a brief with its evidence attached.



Actual local application · v3.0.2 · Local evidence mode

CONFIRM

Task, player IDs, question and controls agree.

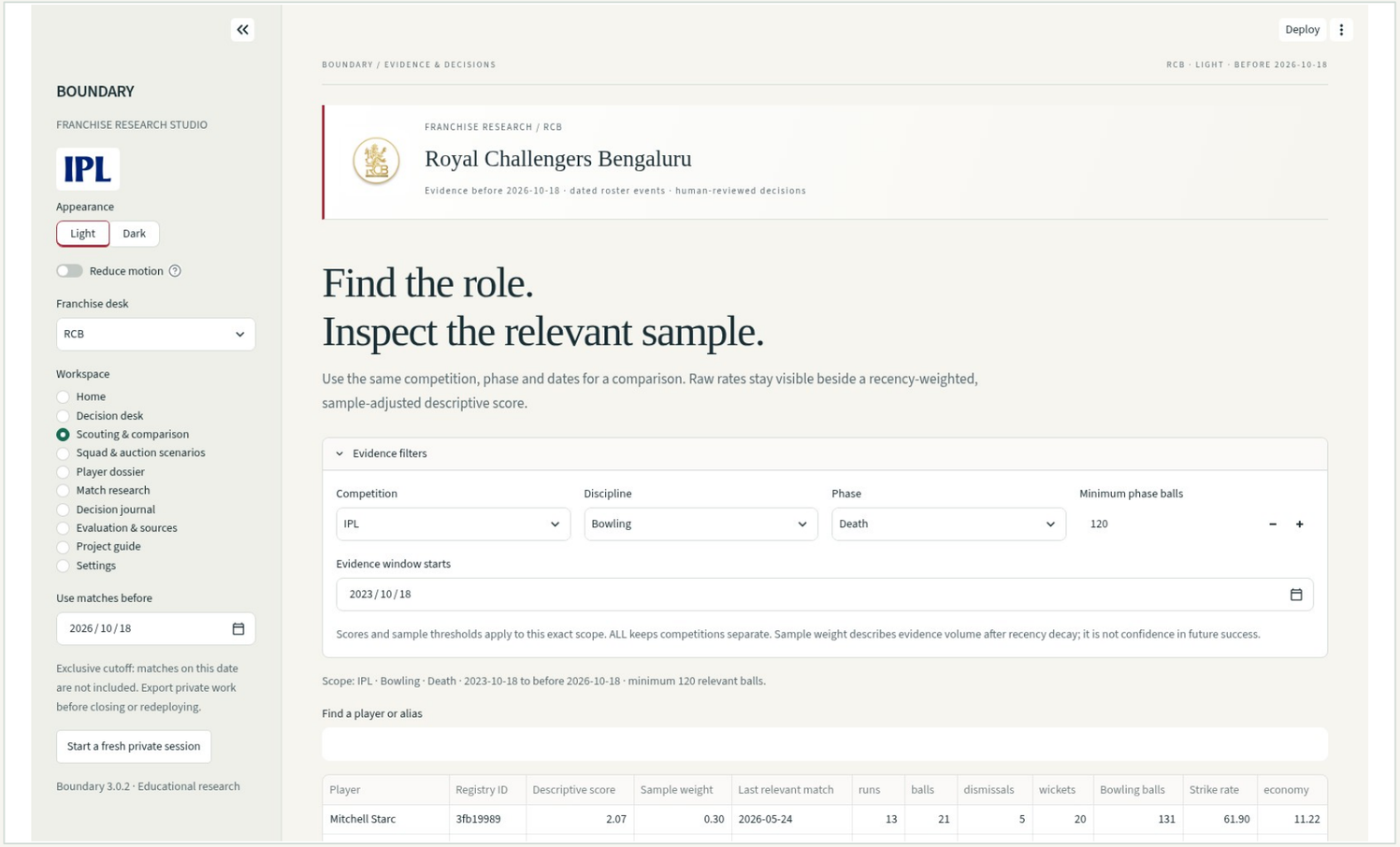
INSPECT

Each fact has a record and evidence trace.

PRESERVE

Save privately or download a decision pack.

Read the raw rate beside the score.



Actual local application · v3.0.2 · Local evidence mode

CONTROL

Competition, discipline, phase, dates and balls.

COMPARE

Up to three registry-linked identities.

INTERPRET

Recency, shrinkage and observed-rate intervals.

A scope is part of every statistic.

114

Death-phase runs

2023-10-18 to before 2026-10-18

2,073

All phases, same window

2023-10-18 to before 2026-10-18

9,336

All supplied IPL records

Archive through 2026-05-31

2024 IPL	2025 IPL	2026 IPL
741 runs	657 runs	675 runs

Player overview and scoped research views label these different meanings explicitly.

Names help search. IDs identify people.

Identity

- Three registry IDs share “Rohit Sharma”.
- Ambiguous names require an explicit choice.
- Observed teams and competitions help confirmation.

Time

- Only matches strictly before the cutoff count.
- The same window is used for fair comparisons.
- Historical XI pools outside 2026 are unsupported.

A missing identity or match cannot borrow somebody else’s evidence.

Selection and the twenty overs are solved together.

11	20	4	4
Distinct selected players	Overs assigned	Maximum overseas	Maximum overs / bowler

- At least one keeper; active role minima and all locks/exclusions.
- No consecutive overs by one bowler; total and phase workload caps.
- Optional assumed budget; independent validation before display.

A real scenario: MI without Jasprit Bumrah.

POWERPLAY / 1–6						MIDDLE / 7–16										DEATH / 17–20				
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
A	B	A	B	C	A	D	E	C	D	E	D	E	D	E	A	B	C	B	C

A AM Ghazanfar

B Deepak Chahar

C Ashwani Kumar

D Mitchell Santner

E Corbin Bosch

20 overs / 120 legal balls / No adjacent repeats / Each named bowler: 4 overs

Make assumptions visible before solving.

Availability and roles

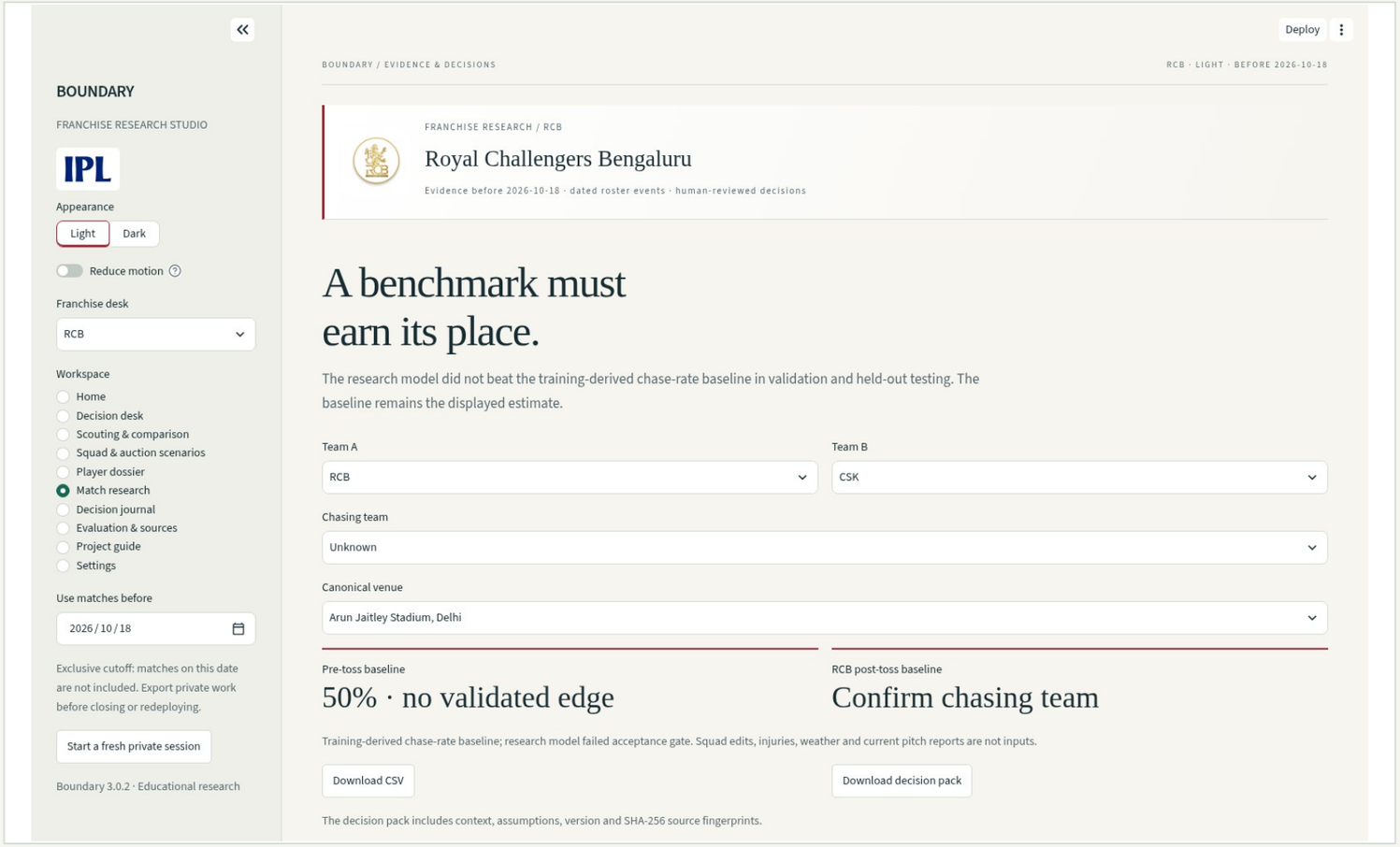
- Exclude unavailable players; lock required players.
- Set supported spin, pace and other role minima.
- Compare a forced alternative and who leaves.

Costs and capabilities

- Enter assumed prices when using a budget.
- Justify any workload or utility override.
- Dated source blocks cannot be overridden.

Historical auction entries do not become live market values.

A model must beat a credible baseline.



Actual local application · v3.0.2 · Local evidence mode

RESEARCH

Regularized logistic model with dated features.

CONTROL

Team A/B, venue and confirmed chasing team.

LIMIT

No lineup, injury, weather or pitch-report input.

Time order prevents future results leaking backward.

BEFORE 2023

Fit fixed model

946 training matches

2023

Validate

73 validation matches

2024 ONWARD

Held-out test

215 test matches

Within each date: forecast every match first, then update ratings from results.

The baseline remains the displayed estimate.

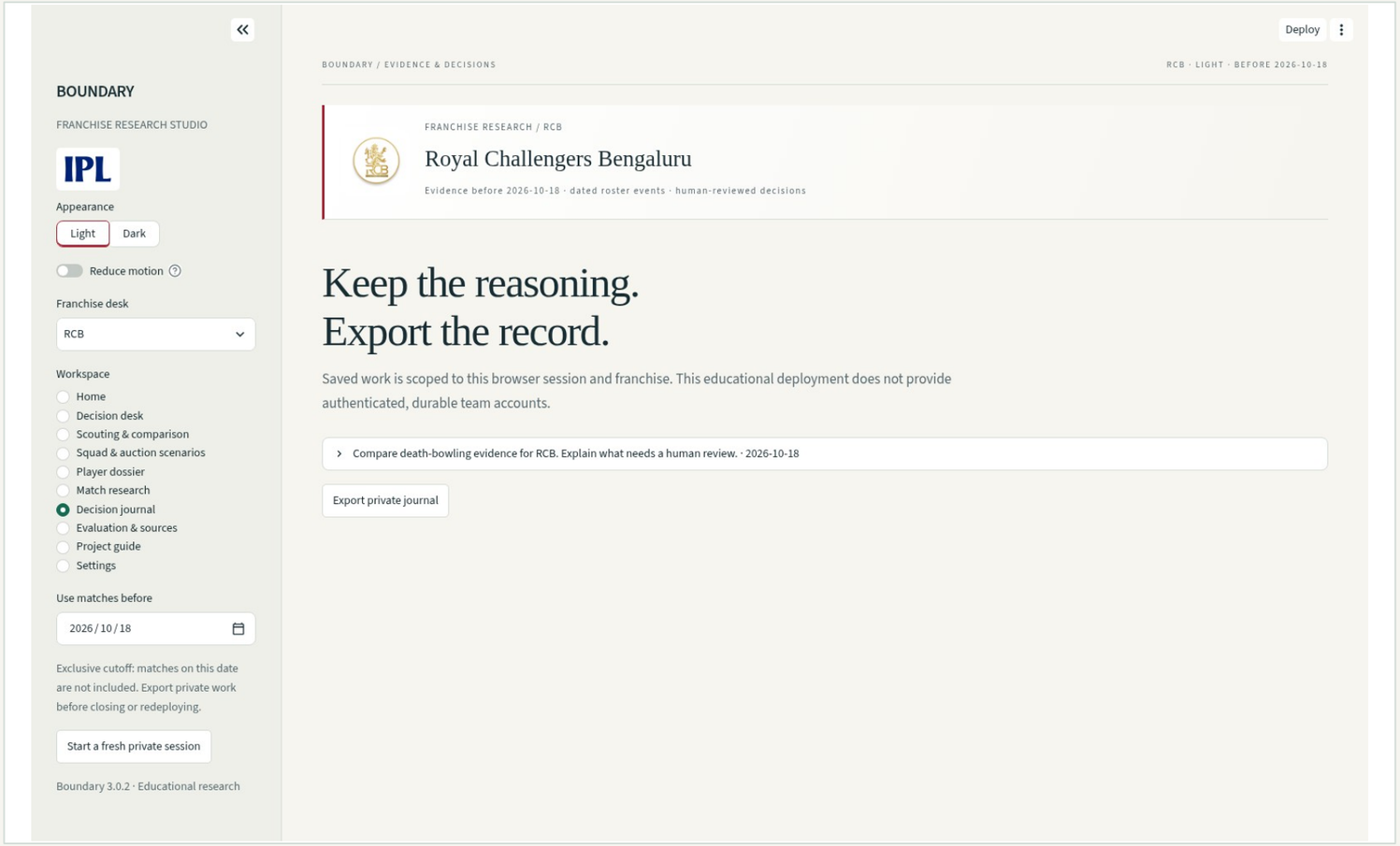
Brier score ↓	Validation (73)	Test (215)
Research post-toss	0.256489	0.248577
Training chase baseline	0.257440	0.247619

GATE: BETTER IN BOTH PERIODS

Not passed

The baseline is served; no predictive edge is claimed.

Keep the reasoning, not just the answer.



Actual local application · v3.0.2 · Local evidence mode

JOURNAL

Saved briefs and scenarios by browser session/team.

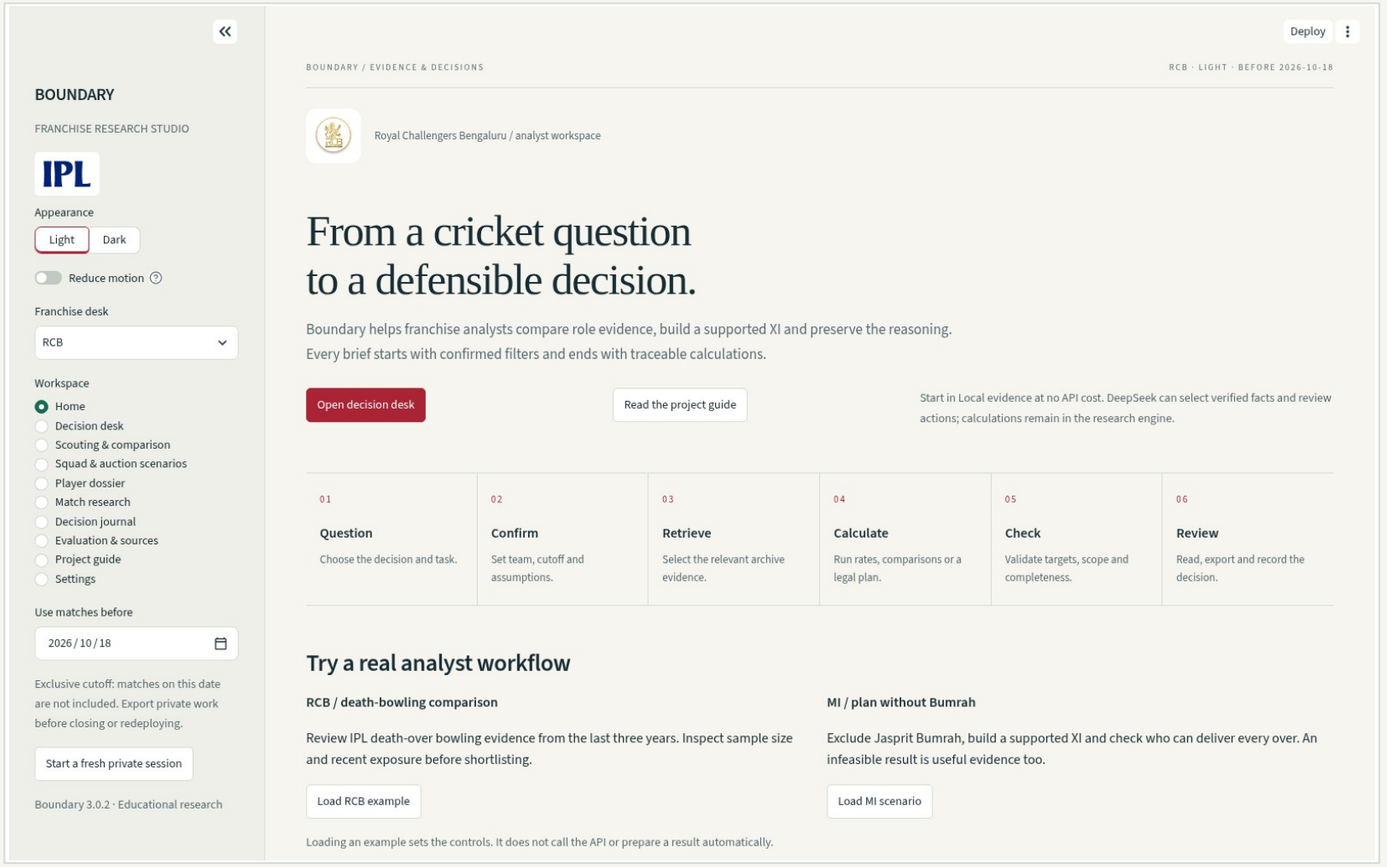
MANIFEST

Question, assumptions, dates and source SHA-256.

EXPORT

CSV, checked evidence and complete XI schedule.

A working studio, with clear visual hierarchy.



CONTEXT

Franchise desk and visible filters

APPEARANCE

Light, Dark and team accents

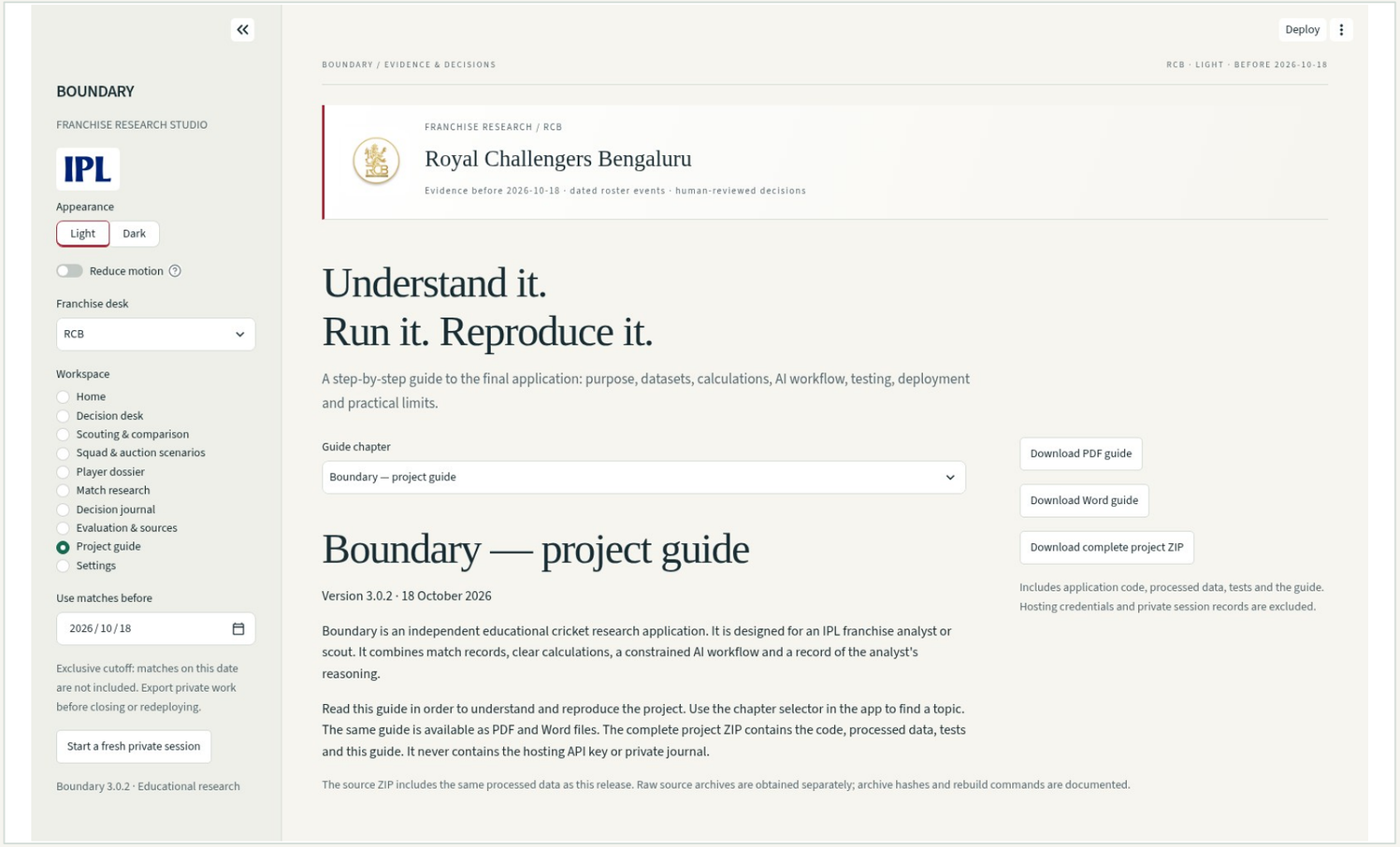
MOTION

Reduced-motion controls

LANGUAGE

Archive scope and uncertainty labels

The application explains how to reproduce it.



Actual local application · v3.0.2 · Local evidence mode

DOCUMENTATION

Same simple-English guide in app, PDF and Word.

SOURCE

Allowlisted project ZIP with data and tests.

SETTINGS

Private key override, connection test and call limits.

Separate functional passes from source completeness.

68

Automated tests

57

Local browser checks

0

Raw-ingestion mismatches

Expanded check	Cases	Interpretation
Scouting settings	144	Competition / phase / date / discipline
All-team scenarios	90	50 supported; 40 explained rejections
Forced-player requests	257	214 supported; 43 explained rejections
Displayed squad dossiers	257	All match stored records or report no identity

Raw reconciliation: 3,393,048 fields across 1,455,366 deliveries. Units are distinct; no combined “accuracy” percentage.

Official comparisons reveal real coverage limits.

188

Matched checkpoints

5

Measured coverage gaps

1

Unresolved conflict

194 numeric checkpoints / 15 primary pages / all ten IPL teams

Kohli T20I	Supplied archive	ICC career reference
Runs before 2024-07-01	3,963	4,188

Source: data/official_validation.csv. This is sampled external validation, not certification of every player.

A supplied pool is not a certified current squad.

257

Supplied pool rows

14

Selected dated notices

7

Missing replacement records

- Block documented outgoing players after publication.
- Block existing replacements before the documented join date.
- Report missing verified metadata; never fabricate a roster.

RCB: shortlist death-bowling evidence.

1	**Load RCB example**	Check IPL · Bowling · Death · 120 balls.
2	**Prepare local brief**	Read economy, wickets, legal balls and last match.
3	**Inspect the evidence**	Compare scope and observed-rate uncertainty.
4	**Save the decision**	Export the brief; list current-availability checks.

MI: remove Bumrah and account for every over.

1	**Load MI scenario**	Bumrah is explicitly excluded in the controls.
2	**Inspect the XI**	Check eleven players, roles and overseas limit.
3	**Open the schedule**	Check every over, phase caps and repeated bowlers.
4	**Test a counterfactual**	Force an alternative or try an impossible budget.

A low-cost capstone, with optional API use.

Local reproduction	Purpose
1. Extract the source ZIP	Processed release data is included.
2. Create a Python environment	Use tested Python 3.12 or a compatible version.
3. pip install -r requirements.txt	Streamlit, pandas, NumPy, SciPy and Altair.
4. python -m streamlit run app.py	Local evidence works without an API key.

Optional hosting + DeepSeek: private Secrets, demo call limits and provider spending controls.

A credible capstone has visible trade-offs.

STRENGTHS

Traceable records

Complete constrained schedule

Reproducible exports

WEAKNESSES

Incomplete frozen coverage

Partial roster metadata

Heuristic score weights

OPPORTUNITIES

Verified dated registry

Video + analyst reviews

Untouched tactical studies

THREATS

Stale availability

Scope misinterpretation

Provider or public-demo abuse

What remains before professional team adoption.

01	**Verify the record**	Fill missing scorecards and resolve primary-source conflicts.
02	**Verify the roster**	Complete dated identities, replacements and availability.
03	**Measure usefulness**	Collect independent analyst ratings and tactical outcomes.
04	**Operate reliably**	Add authenticated storage, monitoring and hosted/API checks.

Evidence.
Constraints.
A decision record.

A practical capstone for inspectable cricket research.

Professional adoption still requires complete data, current roster verification and independent human review.

Questions, demonstration and reproducibility.

B.R. Abishek / AI Driven Data Science

APPLICATION Open the Boundary app

SOURCE br-abishek-07 / boundary-cricket-capstone

FROZEN RELEASE v3.0.2 · 7f891c7

Detailed methods, formulas, safeguards and reproduction commands follow in the appendix.

Batting combines pace and dismissal risk.

$$r^* = (R + 120 \times r_0) / (B + 120)$$

$$d^* = (D + 120 \times d_0) / (B + 120)$$

$$\text{Bat score} = 100(r^* - r_0) - 1{,}800(d^* - d_0)$$

R: weighted runs B: weighted balls D: weighted dismissals

r_0, d_0: competition/phase baseline rates

Declared descriptive weights; no scientifically optimal coefficient or future-value claim.

Bowling combines control and wicket-taking.

$$c^* = (C + 120 \times c_0) / (L + 120)$$

$$w^* = (W + 120 \times w_0) / (L + 120)$$

$$\text{Bowl score} = 6(c_0 - c^*) + 72(w^* - w_0)$$

C: weighted conceded runs L: weighted legal balls W: weighted wickets

c_0, w_0: competition/phase baseline rates

Declared descriptive weights; no scientifically optimal coefficient or future-value claim.

An evidence weight is not a probability of success.

Recency weight = 2^(−age in days / 365)

Sample weight = weighted relevant balls / (weighted relevant balls + 120)

- Raw display totals remain unweighted.
- Observed-rate intervals: 500 match-level resamples, fixed seed.
- At least five relevant matches; independent-match assumption.

The solver’s constraints are explicit.

Object	Constraint
Selection x_i	$\Sigma x_i = 11$; keeper, overseas, role, lock and exclusion rules
Assignment y_{io}	$\Sigma_i y_{io} = 1$ for each over; $y_{io} \le x_i$
Workload	$\Sigma_o y_{io} \le \text{capacity}_i \le 4$; corresponding phase caps
Consecutive overs	$y_{io} + y_{i,o+1} \le 1$
Assumed budget	$\Sigma\ \text{price}_i \times x_i \le$ budget, when enabled

Stage 1: maximize Σ $\text{utility}_i \times x_i$. Stage 2: improve the schedule for the fixed XI.

The schema keeps evidence and assumptions apart.

Input / artifact	Grain and responsibility
match_players.csv.gz	One row per match ID + player ID; full and phase totals
matches.csv.gz	Match dates, competition, teams, venue, toss and result
player_registry.json	Names, aliases, IDs and selected reviewed metadata
squads + dated events/status	Research pool and sourced planning restrictions
model / audit / benchmark artifacts	Evaluation outputs separated from source evidence

A small stack with clear responsibilities.

File / module	Responsibility
app.py + boundary_runtime.py	Streamlit workflow + coherent versioned runtime loading
analytics.py	Scouting, workload support, constrained XI + validation
agent.py	Task checks, permitted tools, fact IDs, provider selection
prediction.py	Chronological research model and baseline serving
evidence.py + storage.py	Context/source fingerprints; scoped journal/cache/allowance
branding.py + downloads.py	Franchise appearance; allowlisted source download

Reject unsupported claims; preserve checked evidence.

Before a provider call

- Reject conflicting controls and unsupported attributes.
- Reserve request allowance atomically.
- Send only the question, context and supplied fact IDs.

After the response

- Validate JSON, IDs and required completeness.
- Reject model prose, tools and unsupported values.
- Use a labeled local fallback when selection fails.

The evidence can be rerun.

Core regression suite

```
python -m unittest discover -s tests -v
```

Expanded workflow + all-dossier audit

```
python tools/check_workflows.py --output work/audit
python tools/check_all_dossiers.py --output work/audit
```

Official numeric sample (dated transcription)

```
python tools/compare_official_checkpoints.py --output work/audit
```

Raw reconciliation (exact five archives needed)

```
python tools/verify_raw.py --source /path/to/raw-data
```

Trace the methods and the source claims.

01 Released implementation and methods

02 Cricsheet archive source

03 Primary numerical checkpoints

04 Data and asset provenance

05 Independent accuracy audit

06 Technology references

07 Optimization implementation reference

All source URLs remain clickable. Team-mark credits: assets/sources.json. No league endorsement is implied.

What to hand in.

Submit	Purpose
Presentation PPTX + PDF	Editable viva deck and portable viewing copy
Project report DOCX + PDF	Academic explanation, methods, results and references
Frozen source ZIP	Runnable code, processed data, tests and guides
User guide + evidence ZIP	Reproduction instructions and completed validation
Abstract + demo/viva guide	Short overview, live demonstration and examiner answers
START_HERE + links/checklist	File map, setup, integrity hashes and institution additions